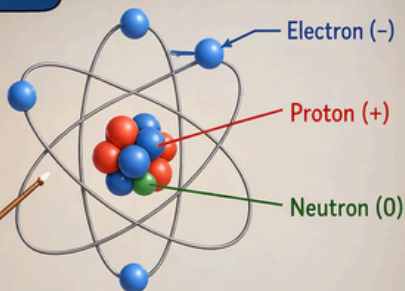
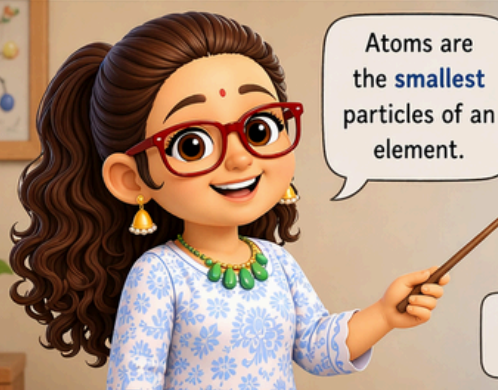


Srijju's Fun Chemistry Lab!

Discover!
Experiment!
Learn!
Have Fun!





Protons and neutrons are in the **nucleus**.
Electrons move around the nucleus.

Part	Charge
Proton	Positive (+)
Electron	Negative (-)
Neutron	No charge (0)

A dot represents millions and millions of atoms!

Hydrogen (1 proton)

Carbon (6 protons)

Oxygen (8 protons)

1 6 8

Hydrogen Carbon Oxygen

The number of protons tells the **identity** of an element.

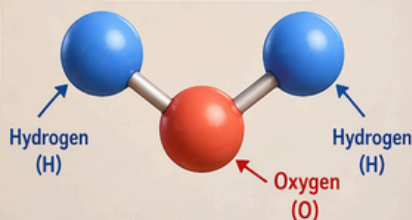
- ✓ Atoms are tiny building blocks.
- ✓ They have **protons, neutrons, and electrons**.
- ✓ Same element = **same** number of protons.
- ✓ Different element = **different** number of protons.

Quick Fact!

Most of an atom is empty space.
The nucleus is very tiny!

When two or more atoms join together, they form a **MOLECULE**.

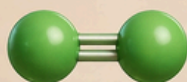
Example: Water (H_2O)



This water molecule has **2 hydrogen** atoms and **1 oxygen** atom.

Molecules can be made of the **same kind** of atoms...

Example: Oxygen (O_2)



Two oxygen atoms join to form O_2 .

...or **different** kinds of atoms.

Example: Carbon Dioxide (CO_2)



One carbon atom and **two oxygen** atoms form CO_2 .

Molecules have a **fixed composition**.



Every water molecule is always H_2O . It doesn't change!

Molecules are **different** from **individual atoms**.

Atom



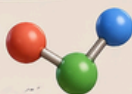
A single hydrogen atom.

Molecule

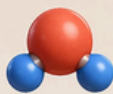


Two hydrogen atoms joined = H_2 molecule.

Let's recap!



Molecules are made of **two or more** atoms.



They can be made of **same** or **different** atoms.



Molecules have a **fixed composition**.



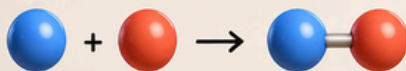
Understanding molecules helps us understand **chemical reactions**!



In short: **Atoms** join together to form **molecules**, the tiny teams behind everything!

Atoms are more stable when they **bond** with other atoms.

Bonds hold atoms together to form molecules and compounds.



Separate atoms

Bonded atoms



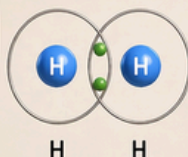
Bonds = the "glue" that holds atoms together.

THERE ARE TWO MAIN TYPES OF BONDS

1 COVALENT BOND

Atoms **share** electrons to fill their outer shells.

Sharing makes atoms happy and stable!



Hydrogen molecule (H_2)

Examples:



Water (H_2O)



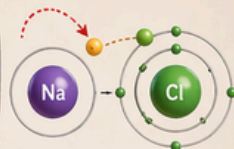
Methane (CH_4)

- ✓ Share electrons
- ✓ Usually between **nonmetals**
- ✓ Forms **molecules**

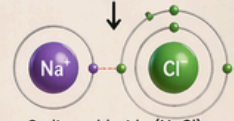
2 IONIC BOND

Atoms **transfer** electrons from one to another.

Transfer creates positive and negative ions!



Sodium (Na) + Chlorine (Cl)



Sodium chloride (NaCl)

Examples:



Table Salt (NaCl)



Magnesium Oxide (MgO)

- ✓ Transfer of electrons
- ✓ Usually between a **metal** and a nonmetal
- ✓ Forms **compounds**

Let's compare!



How electrons are involved



Type of atoms



Forms



Example

Covalent Bond

Shared between atoms

Nonmetal + Nonmetal

Molecules

H_2O , CO_2 , CH_4

Ionic Bond

Transferred from one atom to another

Metal + Nonmetal

Compounds

$NaCl$, MgO , CaF_2

★ Remember:

Atoms bond to become stable. Bonds create the amazing materials all around us!

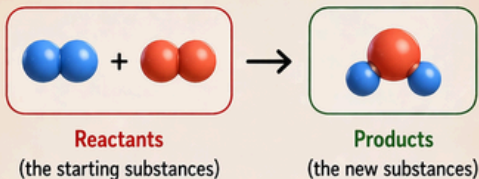


Bonds are the reason tiny atoms can build **big** things!



A chemical reaction is when substances change into **new substances** with different properties.

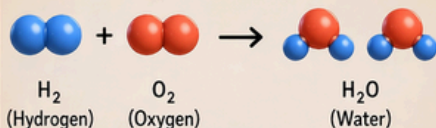
Reactants change and new **products** are formed.



Atoms are not created or destroyed in a reaction. They are only rearranged!

SIMPLE EXAMPLE

Hydrogen + Oxygen → Water

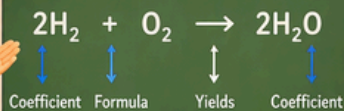


Two hydrogen molecules react with one oxygen molecule to form **two water molecules**.



We show reactions using chemical equations.

Equation:



- ✓ **Coefficients** show how many molecules.
- ✓ **Subscripts** show how many atoms in a molecule.

ENERGY CHANGES IN REACTIONS

Some reactions **release energy** (exothermic). They give out heat or light.



Some reactions **absorb energy** (endothermic). They take in heat from the surroundings.



REACTIONS HAPPEN ALL AROUND US!

Rusting of iron



Iron reacts with oxygen and water to form rust.

Cooking an egg



Heat causes changes in proteins and makes a new substance.

Photosynthesis in plants



Plants use sunlight to turn carbon dioxide and water into food.



Let's remember!

- ✓ Reactants change and form new products.
- ✓ Atoms are rearranged, not lost.
- ✓ Reactions may release or absorb energy.
- ✓ Reactions are a big part of our daily life!



In short: **Chemical reactions** transform substances into **something new!**



Matter exists in **three** main states. They can change from one form to another!



Fun Fact!

Heat can change the state of matter. Cool down or heat up, matter transforms!

SOLID



Has a fixed shape and fixed volume.

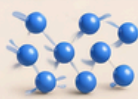


Particles are close together in a fixed arrangement.

LIQUID



Has no fixed shape but has a fixed volume.



Particles are close together but can move around.

GAS



Has no fixed shape and no fixed volume.



Particles are far apart and move freely in all directions.

MATTER CAN CHANGE ITS STATE!



Matter is always on the move! States can change with a change in temperature.



OUR CHEMISTRY JOURNEY – LET'S RECAP!

1. ATOMS



Tiny building blocks of matter.

2. MOLECULES



Atoms join together to form molecules.

3. CHEMICAL BONDS



Bonds hold atoms together to make molecules and compounds.

4. CHEMICAL REACTIONS



Substances change and new substances are formed.

5. STATES OF MATTER



Matter exists as solid, liquid, or gas and can change from one state to another.

CHEMISTRY IS ALL AROUND US!



Cooking



Powering devices



Cleaning



Plants growing



Medicines



Fireworks



Chemistry helps us understand the world and create a **better future!**

YOU DID IT!

Great job exploring the amazing world of Chemistry!

Keep asking questions, keep experimenting, and never stop being curious!



In short: **Chemistry** is the study of matter and the changes it goes through. It's everywhere, it's useful, and it's **awesome!**



A Sriju Comic



Curiosity Lives Here!

Stories that smile back at you